Average Rates of Change

Goals: Compute average rates of change, slopes of secant lines, and motivate the idea of limits
Before Class Videos: None today, starting next class

Example 1: I left my house at time $t = 0$. 30 minutes later, I was at UVic which is 12 km away.

a) What was my average speed?

b) Sketch a possible graph

c) How could I approximate my speed at $t = 10$ minutes?

Example 2: What is average rate of change of $f(x) = x^2$ on $[2, 4]$?

Example 3: Find the equation of the line through (x_1, y_1) and (x_2, y_2).

Example 4: Consider the graph of a function $f(x)$. How do I write an equation for the secant line through $(x_1, f(x_1))$ and $(x_2, f(x_2))$?

Big Idea: As x_2 gets closer to x_1 the slope of the secant line approaches the slope of the tangent line at x_1. We need to investigate and define this limiting process more precisely.

Suggested after class problems from APEX: None today, but review the Course Outline on Brightspace